

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

ITUC301: Data Engineering

Practical File – ODD Semester 2026-27

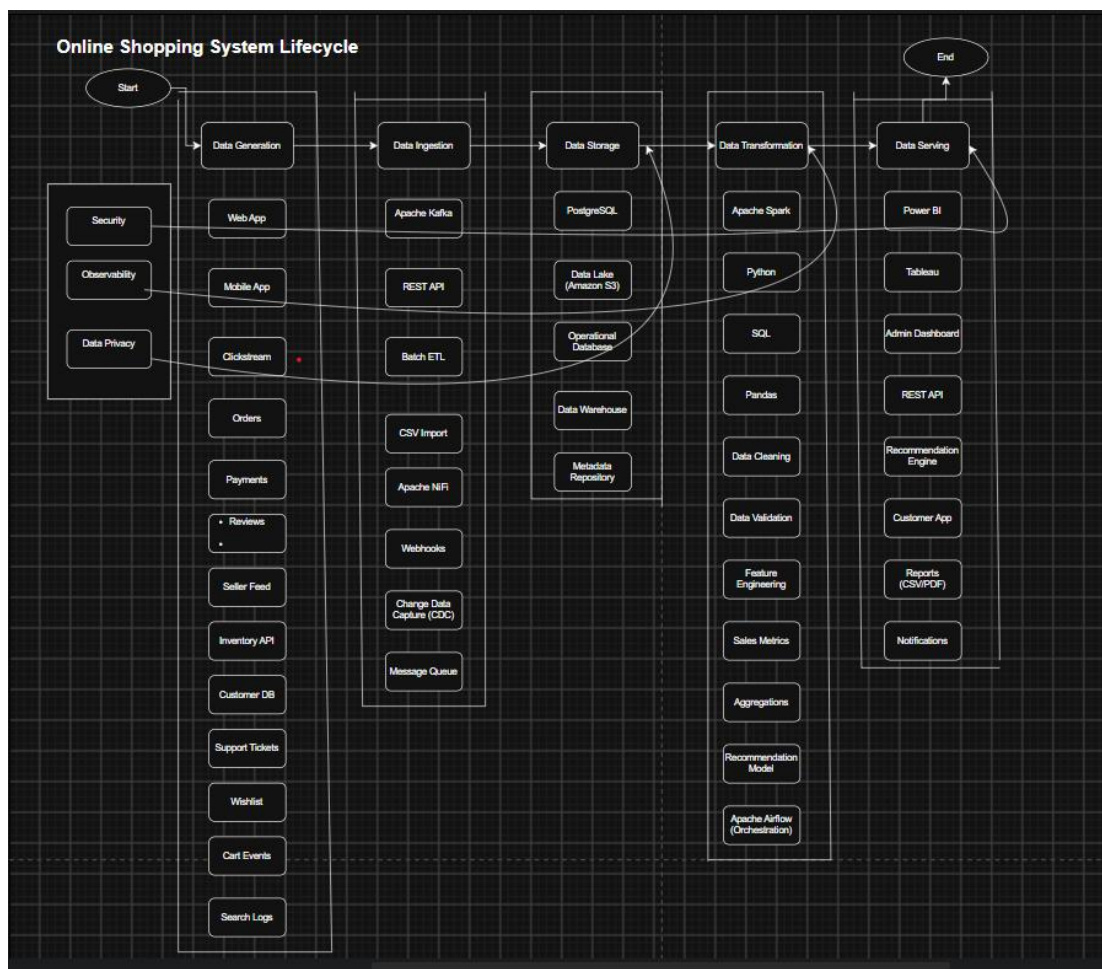
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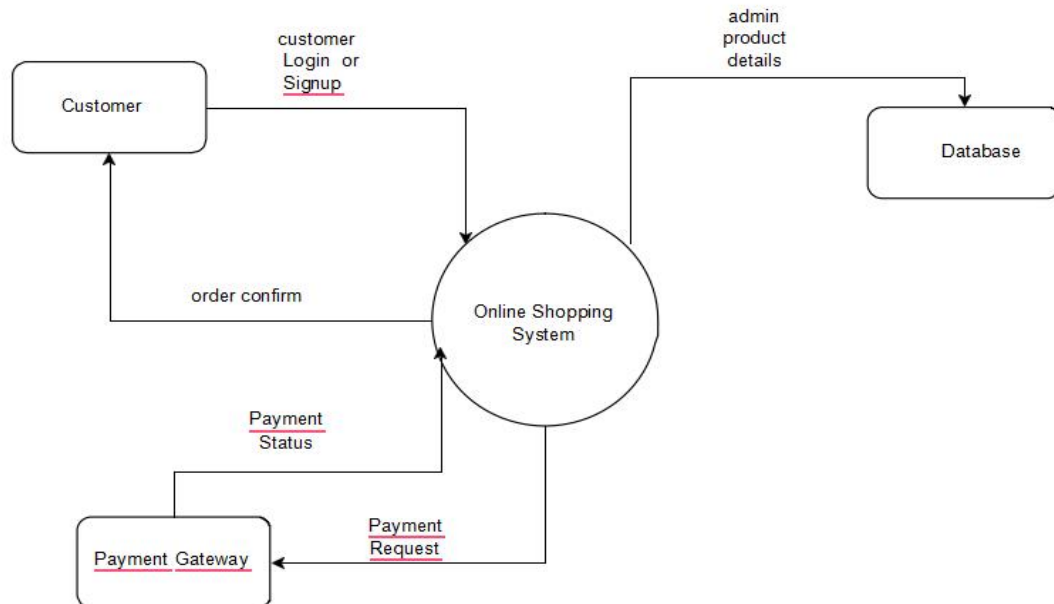
PRACTICAL-1

Practical Definition: Study, dissect, and map the complete end-to-end data engineering lifecycle for an enterprise e-commerce platform. Students must trace data states across all five core lifecycle stages: data generation, ingestion, storage, transformation, and serving. The solution requires creating a detailed architectural data flow diagram along with a system manifesto documenting cross-cutting lifecycle undercurrents (security, observability, and data privacy). This initial structure must be committed to a personal Git repository.

Implementation / Queries and Output/Results



Level 0 Online Shopping System diagram



Key Questions – Answers

- Question:** How do the technical demands of downstream serving layers directly influence ingestion and storage design decisions during the early generation phases?

Answer: The serving layer determines how data will be consumed by business users, dashboards, machine learning models, and APIs. Therefore, during data generation and ingestion, engineers must collect complete, accurate, and structured data. Storage systems are also designed to support fast querying, scalability, and efficient retrieval. If the serving layer requires real-time dashboards, then streaming ingestion and optimized storage are necessary. Thus, downstream requirements significantly influence early lifecycle design decisions.

- Question:** Differentiate between the broader data lifecycle and the technical data engineering lifecycle within an enterprise.

Answer:

Data Lifecycle	Data Engineering Lifecycle
Covers the complete journey of data from creation to deletion.	Covers the complete journey of data from creation to deletion.
Includes governance, compliance, archival, and disposal.	Includes ingestion, storage, transformation, and serving.
Used by the entire organization.	Used mainly by Data Engineers.
Business-oriented.	Technology-oriented

3. **Question:** Identify how the undercurrents of security and data observability should be enforced at each lifecycle boundary.

Answer: Security and observability should be implemented throughout every lifecycle stage.

- **Generation:** Authenticate users and validate input data.
- **Ingestion:** Encrypt data during transfer and monitor pipeline failures.
- **Storage:** Apply role-based access control, encrypt stored data, and maintain backups.
- **Transformation:** Monitor ETL jobs, maintain audit logs, and mask sensitive information.
- **Serving:** Secure APIs with authentication, monitor dashboard performance, and restrict unauthorized access.

Applying these practices ensures data integrity, privacy, compliance, and system reliability.

Conclusion:

The enterprise e-commerce data engineering lifecycle demonstrates how data moves from generation to business consumption through generation, ingestion, storage, transformation, and serving. Cross-cutting concerns such as security, observability, and data privacy are essential at every stage to ensure reliable, secure, and scalable data pipelines. This architecture enables organizations to make accurate, real-time business decisions while maintaining compliance and protecting sensitive customer information.